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CHUNKING TECHNIQUES

That save your **RAG pipeline** in production

Stop using only **fixed chunking + overlap**.
Production RAG needs smarter strategies.



**Broken
Retrieval**



**Smart
Chunking**



**Clean
Pipeline**



AI Coach John

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Fixed / Sliding Window Chunking

📌 WHAT IT IS

Split text into fixed token/character sizes.

Usually:
500 tokens + 50 token overlap.

Most beginners start here.

⚙️ HOW IT WORKS

INPUT

```
Chapter 1: Introduction to AI...
Machine learning allows ...
Deep learning uses neural
networks ...
```



OUTPUT

```
Chunk 1
Chapter 1: Introduction to AI...
Machine learning allows ...

Chunk 2
Machine learning allows ...
Deep learning uses neural
networks ...
```

(Overlap keeps context.)

⚠️ PRODUCTION WARNING

Can split **ideas mid-sentence**.
Semantic meaning may break.

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🎯 WHEN TO USE

- ✓ Clean documentation
- ✓ Short blogs
- ✓ Consistent formatting



Semantic Chunking

WHAT IT IS

Chunk based on **meaning changes**, not token counts.

Sentence embeddings detect topic boundaries.

HOW IT WORKS

INPUT

Python is a programming language.
It supports object oriented programming.

Embedding similarity drops here ↑

Stock markets react to inflation news.
Interest rates affect investments.



OUTPUT

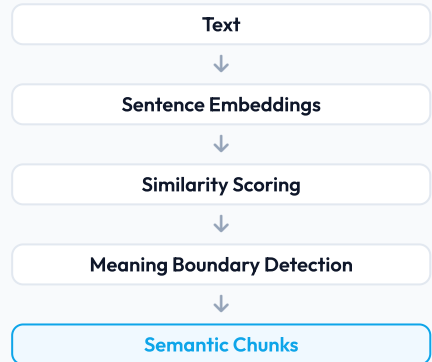
Chunk 1
Python programming section

Chunk 2
Finance section

PRODUCTION ADVANTAGE

Higher retrieval precision.
Less context pollution.

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WHEN TO USE

- ✓ Knowledge bases
- ✓ Research papers
- ✓ Technical docs
- ✓ Multi-topic documents



Hierarchical Chunking

WHAT IT IS

Preserve **document structure**.

Chunk by:

Document → Section → Subsection → Paragraph.

Think tree hierarchy.

HOW IT WORKS

INPUT

```
API Guide
Authentication
├── JWT Tokens
├── OAuth
Rate Limits
├── Requests per minute
```



OUTPUT

```
Level 1 Chunk
Authentication

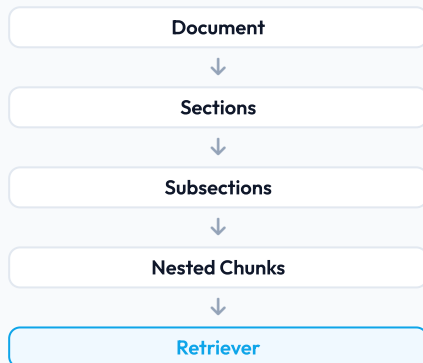
Level 2 Chunk
JWT Tokens

Level 3 Chunk
OAuth
```

PRODUCTION BENEFIT

Maintains parent-child relationships.
Better navigation.

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WHEN TO USE

- ✓ API documentation
- ✓ Enterprise manuals
- ✓ Legal documents
- ✓ Large reports



LLM Based **Chunking**

WHAT IT IS

Let an LLM decide:

"Where should this document be split?"

Useful when data quality is messy.

HOW IT WORKS

INPUT

```
Messy OCR PDF
Invoice#123Paymentterms ...
Section missing...
Broken tables...
```

PROMPT

```
Split this text into logical
knowledge units.
```

OUTPUT

```
Chunk 1
Payment Information
Chunk 2
Terms & Conditions
Chunk 3
Pricing Details
```

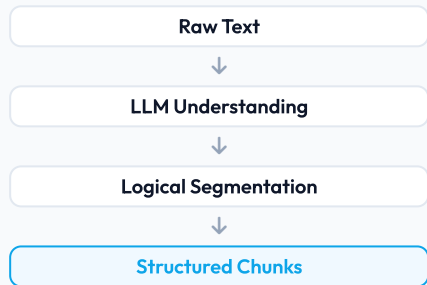
PRODUCTION WARNING

Higher latency.

Higher cost.

Use selectively.

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WHEN TO USE

 OCR data

 PDFs

 Bad formatting

 Mixed layouts



Agentic Chunking

WHAT IT IS

Adaptive chunking.

Agent decides strategy dynamically.
No single splitter.

HOW IT WORKS

INPUT

Mixed Doc [Finance Tables +
Clinical Text]

AGENT REASONING

LLM Analysis:

- Finance data detected (needs precision)
- Health text detected (needs semantic context)

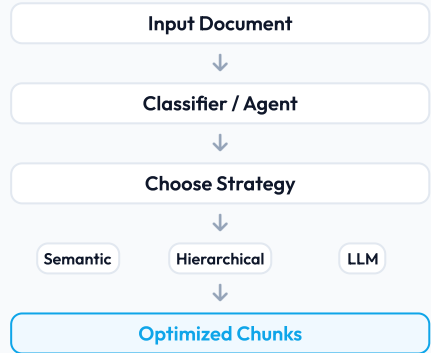
TOOL EXECUTION

Call Tool: Table_Chunker(Finance Segment)
Call Tool: Semantic_Splitter(Health Segment)

PRODUCTION BENEFIT

One pipeline handles many data types.
More scalable.

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WHEN TO USE

- ✓ Enterprise RAG
- ✓ Multi-source ingestion
- ✓ Complex production pipelines





Technique Comparison

TECHNIQUE	BEST FOR	COST	INTELLIGENCE
Fixed	Clean text	\$ Low	Low
Semantic	Topic changes	\$\$ Medium	High
Hierarchical	Structured docs	\$\$ Medium	High
LLM	Messy data	\$\$\$ High	Very High
Agentic	Enterprise systems	Highest	Adaptive



Final Takeaway

Most RAG failures are **retrieval failures**.

And retrieval failures often start with...

bad chunking.

CHOOSE CHUNKING BASED ON:

- ✓ Data structure
- ✓ Document quality
- ✓ Cost constraints
- ✓ Retrieval goals

🔑 Production RAG starts here.



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